Interpretable Adaptive Kernel Banks for Cancer Classification with Limited Data

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Abstract

Cancer classification from annotated medical images is difficult when training data are limited: high-capacity convolutional neural networks (CNNs) can be accurate but require substantial supervision and provide limited insight into the visual evidence used for a decision. We propose an interpretable alternative based on a small bank of jointly learned circularly symmetric kernels. Each image region is represented by pooled responses from the learned kernels, and redundant kernels are identified through correlation of their responses on training data, clustered, sign-aligned, and merged. The resulting model is defined by a small set of one-dimensional radial profiles rather than an opaque filter stack. We evaluate the approach using patient-level splits on a breast TIFF cohort (10,070 regions) and a lower-grade glioma (LGG) MRI cohort (110 patients; 10,000 regions). On breast data, merging reduces a 16-kernel bank to nine kernels without loss of test ROC-AUC (0.9415) and retains 0.8727 accuracy. On LGG MRI, the full bank improves test ROC-AUC from 0.6903 for a single radial kernel to 0.7624; the nine-kernel merged model retains 0.7335. A CNN reference is stronger on breast data (0.9626 ROC-AUC), but the proposed model offers a compact and inspectable representation. Ablation shows that informative pooled responses account for much of the breast improvement, whereas retaining multiple kernels is more important for MRI.

1 Introduction

Medical-image classifiers can support cancer detection, lesion triage, and localization. Their use is especially challenging in limited-data settings, where the number of annotated cases may be too small to reliably train a high-capacity model and where an inspectable account of image evidence is valuable. CNNs have become dominant because they learn useful visual representations directly from data [8, 9]. Their performance, however, is commonly obtained with representations that are difficult to interpret and with data demands that can be impractical for specialized clinical cohorts.

Before end-to-end deep learning, medical-image analysis commonly used explicit visual descriptors: edge, blob, texture, and co-occurrence features [1, 2, 7]. Gaussian derivatives and difference-of-Gaussians describe smoothing, boundaries, and local contrast [3]; Gabor filters capture oriented texture [4]; and histogram-of-oriented-gradients and local binary patterns describe local gradients and intensity relationships [5, 6]. Such representations remain useful in small-data cancer studies, including breast histopathology, mammography, cervical histopathology, and thyroid cytology [13, 14, 15, 16, 17, 18]. Yet prior handcrafted pipelines generally fix a descriptor family in advance. They do not jointly adapt a compact, interpretable kernel set to the data, nor do they explicitly reduce redundancy after learning.

Hybrid approaches offer a useful middle ground: constrained visual features remain inspectable while a lightweight classifier learns the decision boundary. Recent work has combined texture and deep features for renal cancer detection [19], and feed-forward breast-image classifiers have used logistic-regression or SVM modules [20]. Sparse selection [10] and maximal marginal relevance (MMR) [11] can also produce compact feature sets. The outstanding gap is a method that is both adapted to a limited medical dataset and transparent at the level of its learned visual primitives.

We address this gap with a learned radial-kernel bank. Circular symmetry constrains each kernel to a one-dimensional radial profile, reducing its degrees of freedom and making its form directly inspectable. The approach learns several profiles jointly, summarizes their responses with a small interpretable readout, and then merges kernels that are redundant on real training images. The objective is not to replace CNNs universally, but to determine how much discriminative performance can be retained with a compact representation under constrained supervision.

The contributions are:

- a limited-data cancer-classification framework based on jointly learned, circularly symmetric kernels and a lightweight classifier;
- response-space redundancy merging, in which kernels are clustered by real-image response correlation, sign-aligned, and merged into the smallest bank that preserves validation quality;
- an ablation separating the value of multiple kernels from pooled-response encoding, evaluated on breast TIFF and LGG MRI cohorts using patient-level splits; and
- comparison with analytic selected-kernel models, sparse

selection, composite and rotation-invariant variants, and a CNN reference.

2 Methods

2.1 Problem formulation

Let $x_i \in \mathbb{R}^{H\times W}$ be an image region with binary label $y_i \in \{0,1\}$, where positive regions are sampled from a malignant or tumor annotation and negative regions from non-lesion tissue. Given $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, the task is to estimate $p(y = 1 \mid x)$. We use *image region* for the local input to avoid unnecessarily implementation-specific terminology.

For analytic comparison models, kernels are drawn from Gaussian, anisotropic Gaussian, difference-of-Gaussians, Laplacian-of-Gaussian, Gabor, derivative, co-occurrence, local-contrast, and discrete-Laplacian families. Candidate kernels are ranked by univariate ROC-AUC and Fisher score, then MMR selects features that are discriminative and nonredundant. The standard scalar response is

$$r_j(x) = \max_{u,v} |(x * k_j)_{u,v}|, \tag{1}$$

where $*$ denotes convolution. MMR balances relevance against response correlation:

$$k^* = \arg\max_{k_j \notin S} \left[\lambda \mathrm{AUC}(k_j) - (1-\lambda) \max_{k_\ell \in S} |\rho(r_j, r_\ell)|\right], \tag{2}$$

with $\lambda = 0.75$.

2.2 Data and preprocessing

The breast cohort contains high-resolution TIFF images with expert red-contour malignant annotations: 127 malignant images, 365 normal images, and 125 annotation files. Contours are converted to binary lesion masks using color thresholding and morphological cleanup. We extract 128×128-pixel image regions at native resolution. Positive regions require at least 0.80 lesion coverage and negatives at most 0.02; near- and far-lesion negatives are included. Low-tissue regions are excluded.

The LGG MRI cohort of Buda et al. [12], derived from TCGA/TCIA, comprises paired FLAIR slices and manual tumor masks from 110 patients. From the masks, we sample 64×64-pixel regions: positives have at least 0.80 tumor coverage, while negatives have zero coverage, with a subset sampled near the tumor boundary. A tissue-occupancy requirement removes background-air regions. Patient or image identifiers determine fixed 70/15/15 train/validation/test splits, preventing source-image leakage (Table 1). Breast images are not intensity-standardized because absolute contrast is informative; MRI regions are standardized because scanner-dependent offsets are not diagnostically meaningful.

ROC-AUC is the primary metric because it is threshold independent and remains informative under the modest breast-class imbalance. Accuracy is also reported at a threshold selected only on validation data. CNN experiments additionally record precision, recall, F1, loss, and confusion matrices. This separation between model selection and held-out reporting is used for all comparisons, including the choice of the merged-bank size.

Table 1: Image-region counts. Splits are assigned at patient/image level.

Cohort	Split	Pos.	Neg.	Total
Breast	Train	2,900	3,430	6,330
	Validation	1,080	790	1,870
	Test	1,040	830	1,870
	Total	5,020	5,050	10,070
Brain LGG	Train	3,000	3,000	6,000
	Validation	1,000	1,000	2,000
	Test	1,000	1,000	2,000
	Total	5,000	5,000	10,000

2.3 Learned radial kernels and response encoding

The primary model replaces an unconstrained $s \times s$ kernel with a circularly symmetric profile. Let B_r denote the indicator for the shell at distance r from the kernel center. A radial kernel is

$$k = \sum_{r=0}^{R} f(r) B_r, \tag{3}$$

where $f(r)$ is a learned one-dimensional profile. This constraint reduces complexity, preserves an isotropic bias appropriate for blob-like or halo-like patterns, and allows direct inspection. Shells beyond $R_{\max}$ remain fixed at seed values, limiting effective support.

We jointly learn $K = 16$ profiles seeded from rotation-invariant composite profiles and synthetic center–surround shapes. Training uses class-balanced binary cross-entropy through a 32-unit hidden head with dropout 0.1 and profile regularization:

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\mathrm{cls}} &+ \lambda_s \mathcal{L}_{\mathrm{smooth}} + \lambda_c \mathcal{L}_{\mathrm{curv}} \\ &+ \lambda_a \mathcal{L}_{\mathrm{anchor}} + \lambda_n \mathcal{L}_{\mathrm{scale}} + \lambda_d \mathcal{L}_{\mathrm{div}}.\end{aligned} \tag{4}$$

Smoothness and curvature penalize abrupt profile changes; anchoring limits drift from the seed; scale normalization keeps profiles comparable; and $\mathcal{L}_{\mathrm{div}} = \operatorname{mean} \exp(-\|f_i - f_j\|^2)$ discourages collapsed profiles. Checkpoints are selected by validation ROC-AUC, after which a class-balanced logistic model is refit on frozen features.

For response map $z = x * k_j$, each kernel contributes mean absolute response, maximum absolute response, and signed top-m response:

$$\psi_j(x) = \left[\operatorname{mean}|z|,\ \max|z|,\ \frac{1}{m}\sum_{\mathrm{top}-m} z - \frac{1}{m}\sum_{\mathrm{top}-m} (-z)\right]. \tag{5}$$

The absolute summaries capture distributed and localized activity, while the signed term distinguishes opposite contrast polarity. Both absolute summaries are retained because they describe different evidence, not simply because they were previously implemented. A bank of K kernels gives $3K$ features.

2.4 Redundancy merging and implementation

Visually different profiles need not provide independent information on observed data. For normalized response vectors, zero correlation corresponds to orthogonality in response space; high absolute correlation indicates nearly the same direction of evidence. We therefore start with the K-kernel bank and successively consider $K-1$, $K-2, \ldots,$ one-kernel banks rather than assuming the initial count is necessary.

Pearson correlation ρ_{ij} is computed from convolutional responses over a fixed training subset. Kernels are clustered by average-linkage distance $1 - |\rho_{ij}|$. Absolute correlation treats a sign-reversed pair as redundant when its downstream evidence is equivalent. Within each cluster, profiles are aligned to the medoid sign, averaged, and renormalized to the mean member norm; Eq. 3 keeps the result radial. For every achievable count K', the logistic head is refit. We retain the smallest bank whose validation ROC-AUC and threshold-tuned accuracy are within $\tau = 0.005$ of the full-bank values. Test data are used only for final reporting.

The breast configuration uses 31×31 kernels, $R_{\max} = 20$, 25 epochs, and a profile learning rate selected from $\{10^{-2}, 2 \times 10^{-3}, 5 \times 10^{-4}\}$. The MRI configuration uses 63×63 kernels, $R_{\max} = 30$, and 40 epochs. Both use Adam, batch size 128, and validation-only threshold selection. Analytic-bank experiments use 31×31 kernels, 60 epochs, batch size 64, and learning rate 5×10^{-4}. The CNN reference uses 128×128 RGB breast inputs, four convolutional blocks, batch normalization, ReLU, pooling, dropout, adaptive pooling, and a two-layer head; AdamW uses learning rate 3×10^{-4}, weight decay 10^{-4}, patience six, and at most 25 epochs.

The analytic and learned-radial experiments answer complementary questions. Analytic banks test whether a broad library of known visual patterns, selected after response ranking, is sufficient. The learned bank tests whether a strongly constrained representation can adapt to the cohort without becoming an unconstrained CNN. Composite and rotation-averaged models evaluate two alternative ways of simplifying selected filters; ABESS evaluates sparse selection under a different selection objective. Thus, these variants are comparisons of representation and compression strategies rather than interchangeable estimates of one model's uncertainty.

3 Results

3.1 Primary comparison

Table 2 reports held-out breast results. The nine-kernel merged radial bank achieves 0.9415 test ROC-AUC and 0.8727 test accuracy. It exceeds the single radial and analytic selected-kernel baselines while using far fewer kernels than the 90-kernel analytic banks. The CNN reference has the strongest reported performance.

The difference between validation and test values is also informative. The full and merged breast banks have very similar values on both splits, indicating that merging did not simply exploit a favorable validation operating point in this cohort. The CNN's higher test ROC-AUC shows that hierarchical learned features retain additional useful information. At the same time, the radial-bank result demonstrates that much of the discriminative signal can be represented using a small number of constrained local patterns, which is the intended operating regime for limited-data, inspection-oriented use.

3.2 Minimal-bank results and ablation

The full radial bank improves over its single-kernel counterpart on both cohorts (Table 3). On breast data, merging reduces 16 profiles to nine without a measurable test loss. On MRI, the full bank also improves markedly over a single kernel, but the validation-selected nine-kernel bank loses some performance on test; this indicates that one validation split may be insufficient to certify a merge in a noisier cohort.

The merge sweep in Table 4 separates kernel count from encoding. On breast data, even the merged one-kernel configuration reaches 0.9346 test ROC-AUC, above the 0.9212 obtained by the directly optimized single-kernel reference. Since the former uses Eq. 5 and the latter uses only Eq. 1, much of the improvement follows from preserving complementary response information. Increasing from one to nine merged kernels gives a further but smaller breast gain. On MRI, performance falls substantially as K' approaches one, indicating that multiple spatial patterns are important for this modality.

Intensity preprocessing is modality dependent. Applying per-region standardization to the breast configuration reduces test ROC-AUC from 0.9411 to 0.8605 and accuracy from 0.8754 to 0.7898.

The two datasets therefore provide complementary evidence rather than a single pooled estimate. Breast images reward an encoding that preserves absolute intensity and gain relatively little after the first well-encoded radial profile. MRI requires standardized inputs and benefits more from retaining distinct profiles. This modality dependence cautions against a universal preprocessing recipe or a fixed kernel count: both should be treated as validation-selected design choices.

Table 2: Representative held-out breast results.

Model / configuration	Features	Val AUC	Val Acc.	Test AUC	Test Acc.
Minimal merged radial bank	**9**	**0.9373**	**0.8856**	**0.9415**	**0.8727**
Radial kernel bank before merging	16	0.9388	0.8861	0.9411	0.8754
Learned single radial kernel	1	0.9189	0.8545	0.9286	0.8668
Patch CNN reference	learned	0.9445	0.8818	**0.9626**	**0.9021**
Selected kernel bank with QMC sampling	90	0.9270	0.8603	0.9387	0.8866
Selected kernel bank, variant A	90	0.9271	0.8636	0.9377	0.8882
Sparse ABESS-selected bank	200	0.9095	0.7849	0.9236	0.7690
Shape-sensitive triangular/square bank	24	0.9109	0.8481	0.9221	0.8465

Table 3: Radial banks and single radial kernels. Merged banks are selected on validation data; accuracy uses a validation-tuned threshold.

Cohort	Model	Kernels	Features	Val AUC	Test AUC	Test Acc.
Breast	Best single learned radial kernel	1	1	0.9100	0.9212	0.8412
	Radial kernel bank	16	48	**0.9388**	0.9411	**0.8749**
	Minimal merged bank	**9**	**27**	0.9373	**0.9415**	0.8733
Brain LGG	Single learned radial kernel	1	3	0.7281	0.6903	0.6365
	Radial kernel bank	16	48	**0.7849**	**0.7624**	**0.6750**
	Minimal merged bank	**9**	**27**	0.7801	0.7335	0.6640

Table 4: Representative merge-sweep ROC-AUC values. Asterisks denote validation-selected banks.

	Breast		Brain LGG	
K'	Val	Test	Val	Test
16	0.9389	0.9411	0.7850	0.7624
11	0.9364	0.9407	0.7755	0.7531
9*	0.9373	0.9415	0.7801	0.7335
6	0.9299	0.9356	0.7593	0.7256
2	0.9259	0.9326	0.7039	0.6964
1	0.9271	0.9346	0.6218	0.6671

3.3 Additional comparisons and qualitative assessment

Compact MMR models confirm that a single analytic kernel can be discriminative: reported test ROC-AUC ranges from 0.9157 to 0.9239 for one- to five-kernel configurations. Their accuracy varies more than their AUC, suggesting that univariate ranking does not guarantee a stable operating threshold. Sobol quasi-Monte Carlo sampling improves a matched 90-kernel selected-bank model from 0.9195 to 0.9387 test ROC-AUC and from 0.8646 to 0.8866 accuracy. In contrast, composing selected kernels by normalized summation loses information: 90-kernel models with test ROC-AUC of 0.9376–0.9387 fall to 0.8934–0.9026 when represented by eight composites. Rotation averaging provides exact right-angle invariance but has a reported cost (0.8755 versus 0.8933 test ROC-AUC for the corresponding non-invariant composite). ABESS selection is competitive in AUC (0.9236 for 200 features) but has lower reported accuracy (0.7690).

Figures 1–3 provide diagnostic outputs: confusion matrices and probability overlays on held-out images. They allow inspection of whether high-response areas correspond to lesion-like tissue, but no expert localization scoring or pixel-level localization metric is claimed.

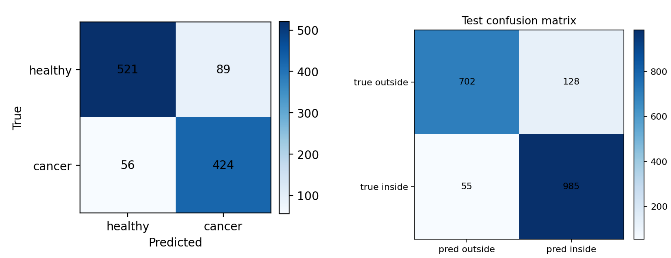

Figure 1: Confusion matrices for the selected-kernel classifier (left) and CNN reference (right).

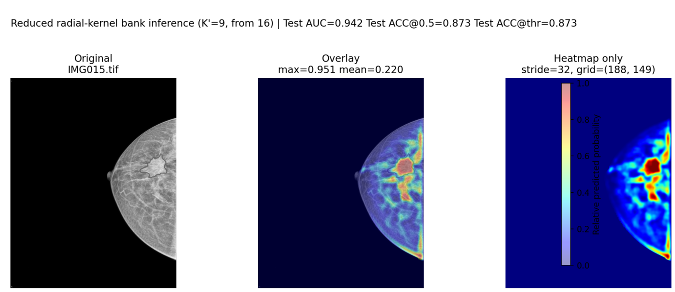

Figure 2: Held-out breast-image probability overlay from the minimal merged radial bank.

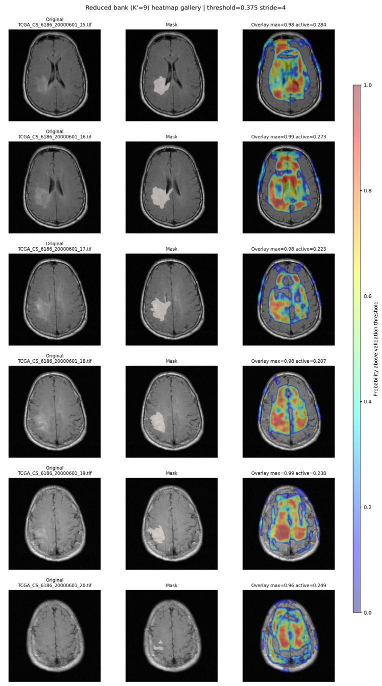

Figure 3: Held-out LGG MRI slices: image, tumor mask, and minimal-bank probability overlay.

4 Discussion and Conclusion

This study addresses limited-data cancer classification with an interpretable representation whose visual primitives remain explicit. The central objective was achieved on the available cohorts: jointly learned radial profiles, followed by response-space merging, retain strong discrimination with a small number of kernels. On breast images, nine profiles preserve the performance of a 16-kernel bank and outperform the reported analytic selected-kernel baselines; on MRI, the full bank demonstrates that multiple radial patterns can improve substantially on a single profile. These findings support the contribution that compactness can be obtained by merging evidence-equivalent kernels rather than by fixing a small bank a priori.

The results also clarify where model capacity resides. On breast data, pooled encoding is a major source of improvement because it retains distributed, localized, and contrast-polarity evidence. On MRI, the number of kernels has a larger role, consistent with more variable tumor appearance across spatial scales. The CNN remains the best-performing breast reference, so the proposed method should be understood as an interpretable complement rather than a replacement for deep models where abundant data and opaque representations are acceptable.

Several limitations qualify these conclusions. Evaluation is at the image-region level, not the patient diagnostic level; labels originate from annotation coverage and are ambiguous near boundaries; reported differences have no repeated-seed confidence intervals; and qualitative maps lack formal localization metrics. The merge criterion also relies on one validation split, which did not fully transfer on the MRI cohort. Future work should use repeated or cross-validated merge selection, external clinical cohorts, formal localization evaluation, and a matched CNN reference on MRI.

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